

Parameter-Driven Refinement of a Sea Snake Genome Assembly Enhances Structural Accuracy and Enables Improved Gene Quantification

ABSTRACT

The fidelity of evolutionary and functional inferences in genomics is inherently constrained by the quality of underlying genomic resources; thus, assemblies warrant iterative refinement as novel bioinformatic tools emerge.

Discrepant vomeronasal type-2 receptor (V2R) counts among recently diverged *Hydrophiinae* sea snakes—who's expanded V2R repertoire underpins underwater chemosensory adaptation following terrestrial-to-marine transition—prompted re-evaluation of *Hydrophis major*'s chromosome 2, revealing low coverage, poor mapping quality, pervasive soft-clipping, and syntenic inter-species misalignment with Z (sex) chromosome segments, in V2R-dense regions—challenges recurrent across the subfamily. Systematic parameter optimisation across the bioinformatical pipeline employed sequential quality evaluation using various metrics. Contrary to conventional approaches prioritising contiguity, liberal parameters initially increasing contig fragmentation offered a 'higher ceiling' for assembly quality by preventing premature joins, yielding smaller, more reliable contigs that could be accurately scaffolded later using Hi-C evidence. Subsequent manual Hi-C scaffolding in Juicebox yielded a chromosome-scale assembly with uniform coverage, elevated completeness and reduced duplication and error rates. Re-annotation uncovered 570 V2Rs on chromosome 2—substantially exceeding prior estimates of 300–400—attributable to resolved Z-derived sequences.

These advances affirm that targeted parameter tuning with emerging computational tools enhance genomic completeness, bolstering the reliability of gene-family quantification and evolutionary inference in complex non-model taxa.

Background

The fidelity of evolutionary and functional inferences in genomics is inherently constrained by the quality of underlying genomic resources (Priyam et al. 2021). Assembling complex genomes, particularly non-model taxa with repetitive and structurally variant regions, necessitates iterative refinement, leveraging emerging bioinformatic tools to enhance contiguity, accuracy, and completeness, as “good data alone cannot guarantee a good assembly” (Sutton et al. 2021). These improvements are essential to enable precise mapping of gene families and structural variants implicated in adaptive evolution.

Hydrophiinae, a subfamily within the *Elapidae* family, is comprised of sea snakes predominantly inhabiting the coastal waters of Australia, Southeast Asia, and the Indo-Pacific. This lineage, estimated to have diverged from terrestrial Australian elapids roughly 9–18 million years ago (Ludington et al. 2023), represents a relatively young yet rapid marine radiation. Their transition from land to sea required

substantial physiological and ecological adaptations, including modifications in locomotion, respiration, and sensory systems, making them an excellent model for studying mechanisms underpinning recent adaptive radiations and evolutionary novelty in vertebrates.

Central to the sensory adaptation of *Hydrophiinae* is the vomeronasal organ (VNO), an auxiliary olfactory structure specialised for detecting waterborne chemical cues. The vomeronasal type-2 receptor (V2R) family, expressed in the VNO, encodes G-protein-coupled receptors integral to chemical signal transduction processes (Li et al. 2021). These receptors mediate behaviours essential for foraging, prey discrimination, and reproductive success (Li et al. 2021). The diversity and counts of V2R genes thus bear directly on ecological fitness and evolutionary trajectories of sea snakes (Li et al. 2021). An accurate understanding of V2R repertoire and function demands comprehensive genomic resources, including high-quality assemblies and thorough gene annotation. Prior studies have reported significant disparities in V2R counts across *Hydrophiinae*, highlighting challenges related to incomplete or misassembled genome regions, especially on chromosome 2, known for dense V2R clusters. Previous estimates ranged widely: Ludington & Sanders (2025 (unpublished)) reported 340 V2Rs on chromosome 2 of *Hydrophis major*, contrasting sharply with some species exhibiting under 50 (figure 1). Li et al. (2021) reported approximately 390 chromosome 2 V2Rs in recently diverged species (*H. curtus*), underscoring potential annotation and assembly biases. Scepticism of V2R annotation and quantification additionally stemmed from synteny analyses revealing poor collinearity and assembly accuracy in V2R-rich, repeat dense regions, particularly at the start of chromosome 2 and in the sex (Z) chromosome (figure 2), regions prone to structural variation and complex repeats (Li et al. 2021; Ludington et al. 2023). Such genomic architecture complicates downstream evolutionary interpretation and functional gene repertoires.

Here, these methodological challenges are addressed by employing systematic optimisation of assembly parameters, prioritising reliability over contiguity via liberal initial contig fragmentation, followed by manual scaffolding guided by Hi-C data. Resulting chromosome-scale assemblies exhibited uniform coverage, improved completeness, and fewer artifacts, culminating in an enhanced annotation of V2R genes. This refined assembly revealed 570 V2Rs on chromosome 2 of *H. major*, substantially exceeding previous estimates, partially due to the resolution of Z chromosome-derived sequence ambiguity (Li et al. 2021; Ludington & Sanders 2025 (unpublished)). These improvements not only refine our understanding of the scale and distribution of V2R gene families in this key marine lineage but also emphasise the critical dependency of evolutionary insight on assembly quality. Our findings highlight that targeted parameter tuning, complemented by updating genomic resources with emerging computational strategies, unlocks new potential for gene-family identification and comparative genomics in repetitive, structurally complex genomes, enhancing our grasp of marine sensory adaptation and speciation dynamics (Priyam et al. 2021; Rhie et al. 2020; Zeng et al. 2023).

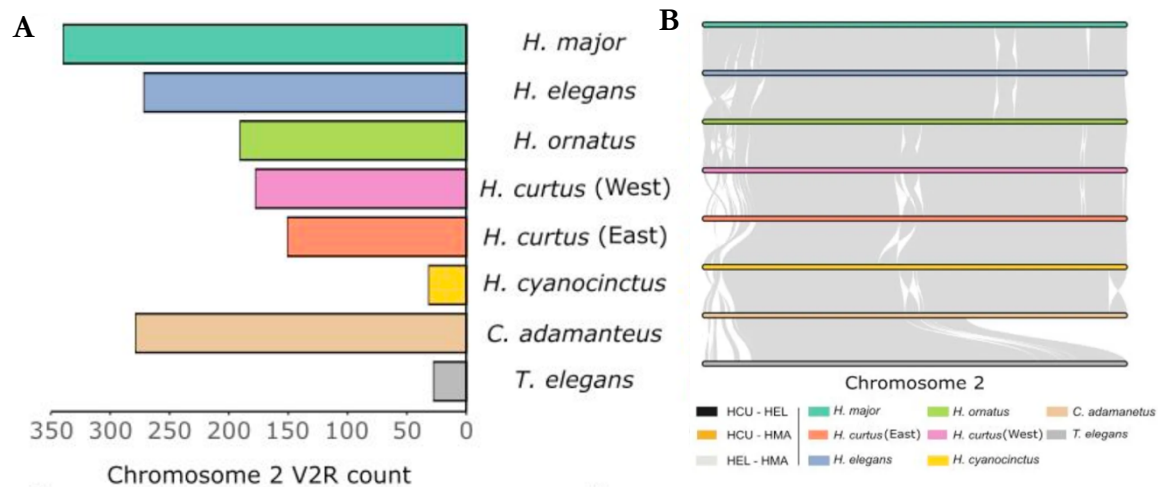
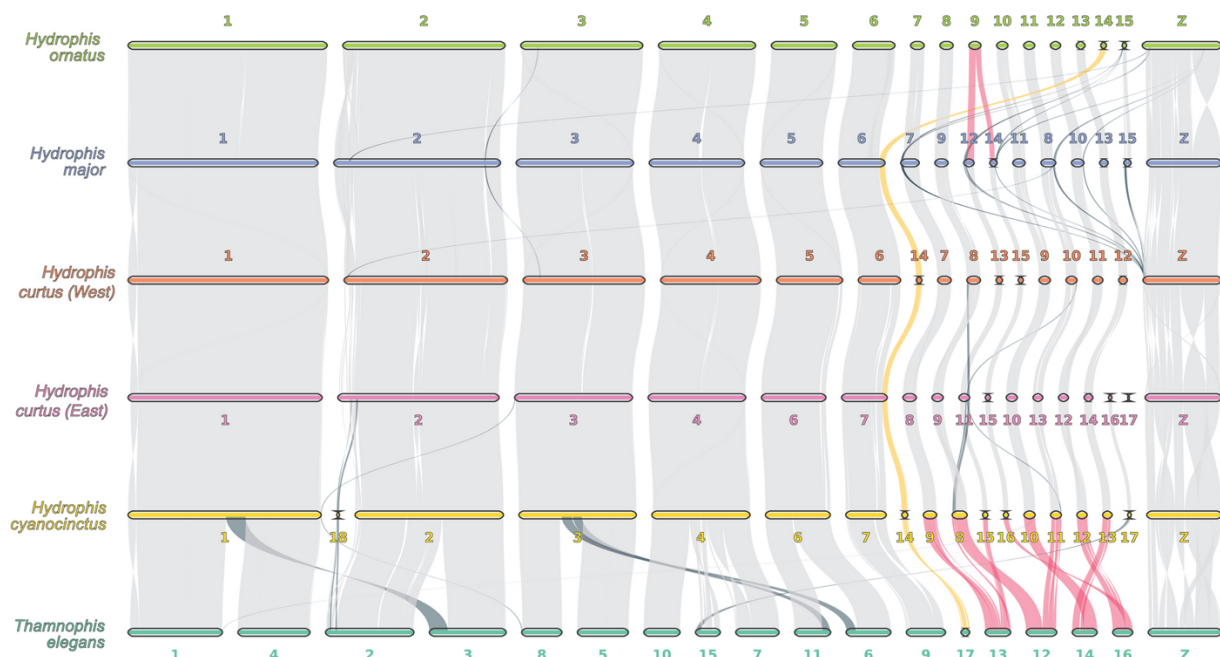


Figure 1 (above) (reproduced from Ludington & Sanders (2025 unpublished): **A**) V2R gene composition along the chromosome 2 in *Hydrophis* species, and *C. adamanteus*, *T. elegans* as terrestrial outgroups for reference. **B**) Chromosome 2 synteny. Note the abundance of unsyntenic structural rearrangement at the start. **Figure 2 (below)**: Chromosome-scale synteny ribbon plot (reproduced from Ludington et al. (2023))



Experimental Procedures & Outcomes

Reference Genome & Coverage Assessment

Custom analysis pipelines and scripts developed for this project, along with supplementary results, are publicly available at <https://github.com/blytrm>. The publicly available *Hydrophis major* genome (GCA_033807585.1; available at https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_033807585.1/) was used to evaluate assembly continuity and completeness. Read coverage was computed with Mosdepth to assess sequencing depth across a subset of chromosome 2 (0–60 Mb) (Pedersen & Quinlan 2018). Coverage irregularities were visualised using R and plotted as binned, mean normalised coverage (figure 3). Regions exhibiting pronounced coverage dips were loaded into Integrative Genomics Viewer (IGV v2.17)

to verify mapping integrity, soft-clipping, supplementary alignments, and mapping quality (MQ) deviations (figure 4) (Robinson et al. 2011). Visual inspection with the genome browser revealed that low-coverage regions along chromosome 2 coincided with clusters of soft-clipped and low-MQ reads, frequently bridging to loci on the Z chromosome (figure 4). This pattern suggested potential misassembly or chimeric contigs linking chromosome 2 and Z chromosome fragments (Rhie et al. 2020). Given that *H. major* females possess a ZW system, it was hypothesised that these misalignments derived from haploid Z-linked regions merged with autosomal contigs during the initial assembly because of low-coverage (figure 5A).

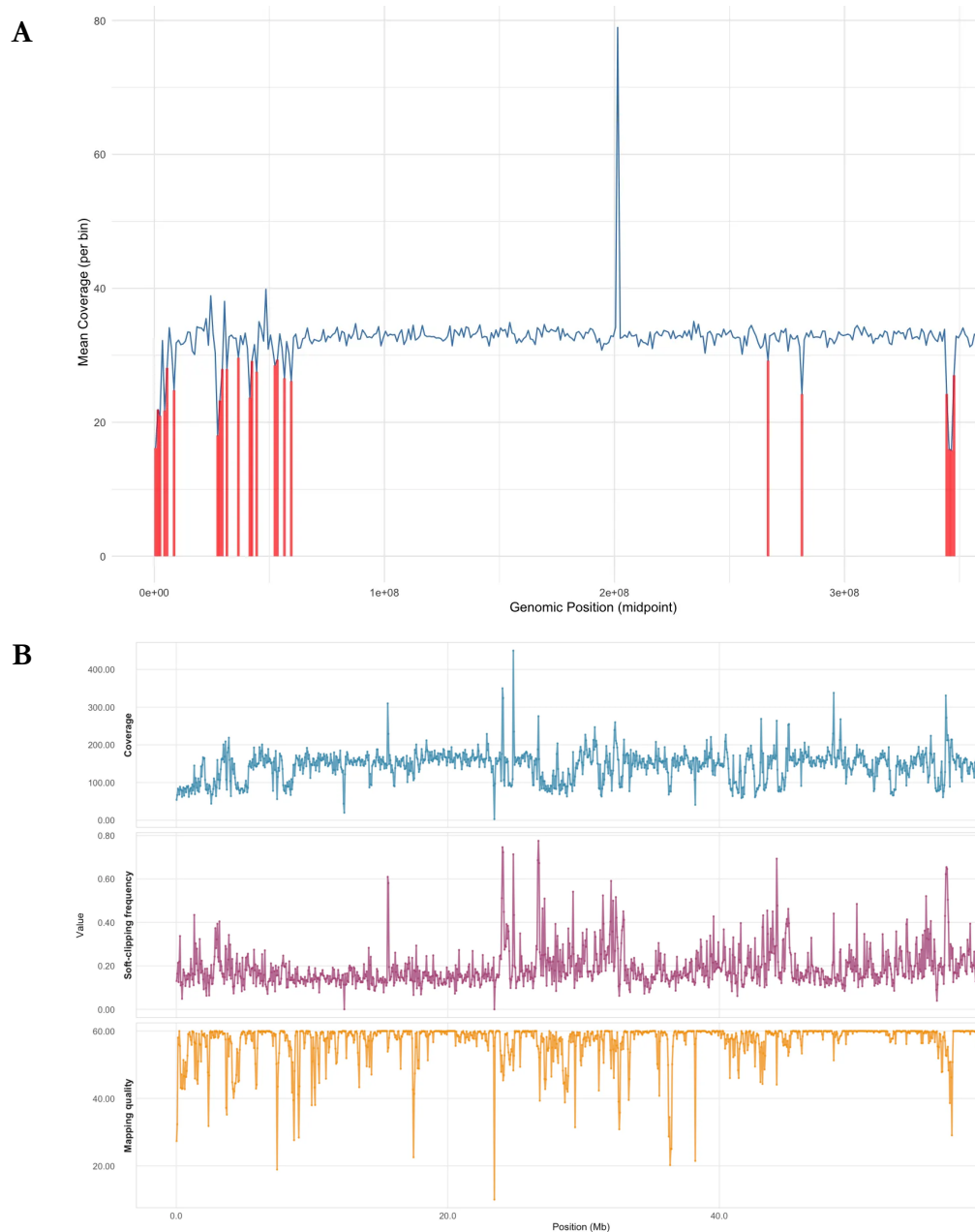


Figure 3 (above): A) Chromosome 2 coverage plot (0-60Mb). Data was binned (1Mb) and normalised (mean). Highlighted red are regions where coverage drops below 30 significantly. **B)** Chromosome 2 quality investigation plot depicting the coverage, soft-clipping frequency and mapping quality at the start (0-60Mb). Note that regions low in coverage generally see a reduction in mapping quality and spike in soft-clipping.

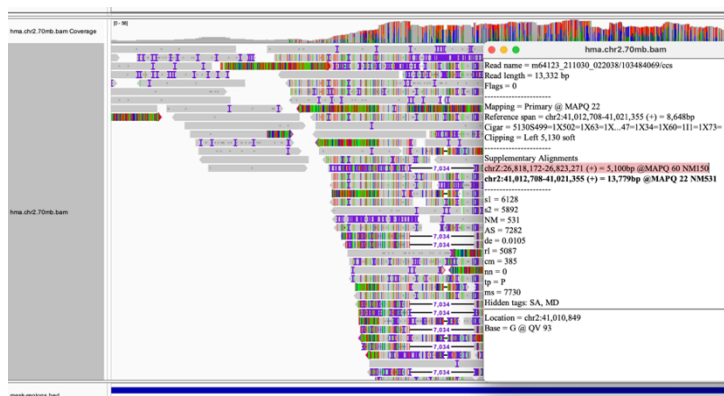


Figure 4 (right): A screenshot taken from IGV genome browser of the low-coverage regions determined from chromosome 2 displaying the abundance of soft-clipping (rainbow-coloured bars) where coverage tends to drop. As well, supplementary alignments to the Z chromosome exist here.

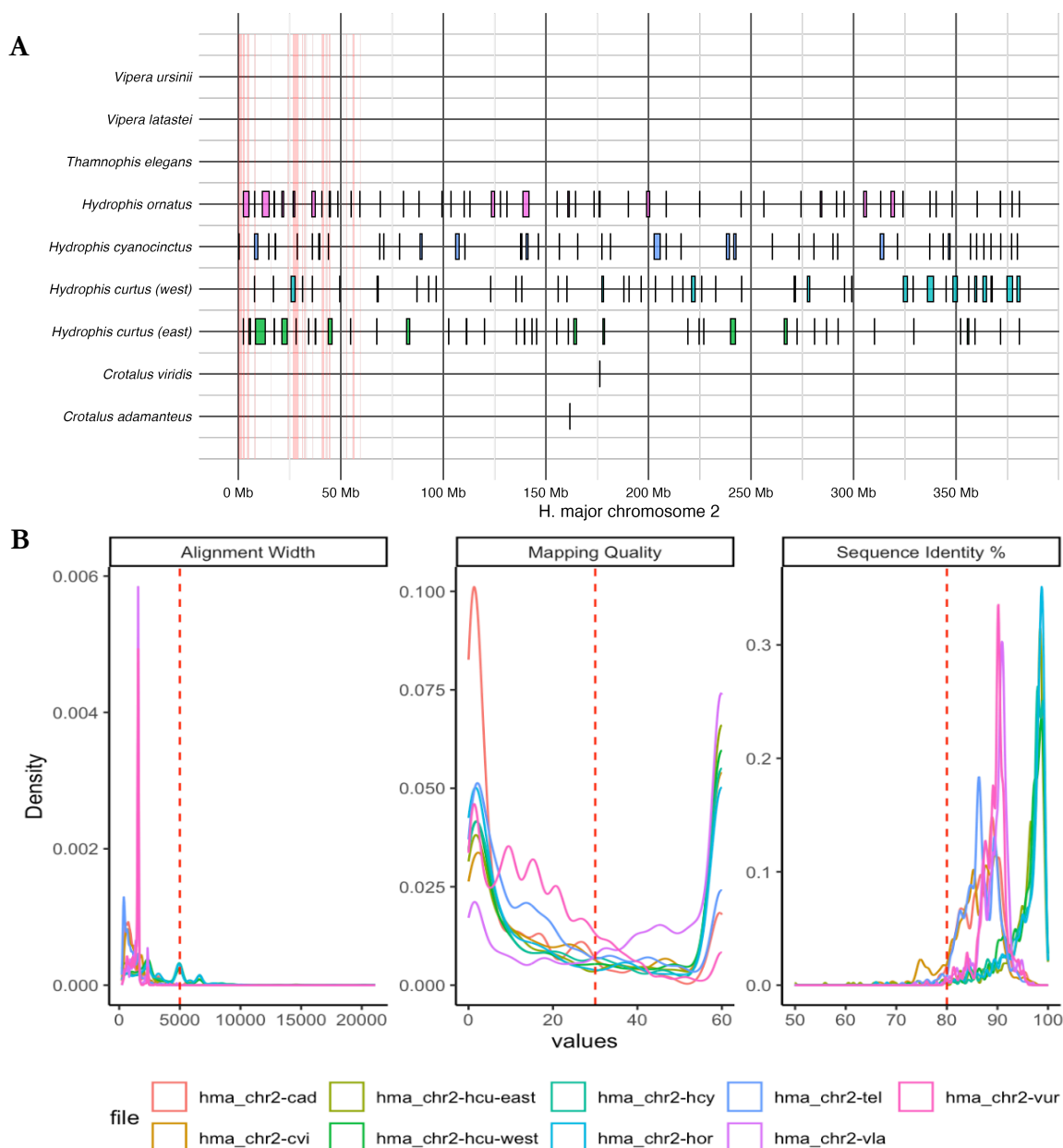


Figure 5 (above): A) Visualisation of pairwise alignment between *H. major*'s chromosome 2 and the Z chromosome of a selection of *Hydrophis* species and terrestrial outgroups. Red lines indicate the low-coverage regions determined previously, which coincides with larger, high-confidence segments. **B)** Density plots of the different quality control parameters, used to determine the final set of parameters. Red dotted lines indicate the values used in the final alignment QC.

Inter-Species Alignment & Quality Control

To test this hypothesis, chromosome 2 (0–60 Mb) from *H. major* was aligned against the Z chromosomes of a selection of snake species (*H. ornatus*, *H. cyanocinctus*, *H. curtus (west)*, *H. curtus (east)*, *Crotalus viridis*, *Crotalus adamanteus*, *Thamnophis elegans*, *Vipera latstei* & *Vipera ursinii*) using Minimap2 (Li, 2018). Alignments were filtered with MAPQ ≥ 30 , alignment length ≥ 2500 bp, and sequence identity $\geq 80\%$. The resulting PAF files were postprocessed in RustyBam to orient and trim alignments (<https://github.com/mrvollger/rustybam>). Parameter thresholds were determined empirically by evaluating the per-parameter distributions (figure 5B), from 0-10Mb subset of chromosome 2, adopting conservative cut-offs to retain only high-confidence cross-chromosomal matches.

Apparent from the visualisation in figure 5A, are the abundance of high-confidence regions of misalignment between *H. major*'s chromosome 2 and inter-species' Z chromosome, areas of which coincide with the original low-coverage locations. Substantiated by these issues in the current genome assembly, it was accordingly decided to re-assemble a revised chromosome-level assembly for *H. major*, to attempt to resolve the challenging regions.

De Novo Genome Reassembly Following Misassembly Detection

Following confirmation of these structural inconsistencies, de novo reassembly was performed using the raw *H. major* long-read dataset to generate an improved haplotype-resolved genome. The updated assembly aimed to resolve the detected chimeric joins and improve scaffold contiguity without introducing artificial concatenations. Reassembly was executed using Hifiasm (Cheng et al. 2021), selected for its demonstrated accuracy in diploid and heterozygous genomes. Six parameter configurations were tested, encompassing combinations of default and alternative inputs, thus enabling systematic evaluation of how post-assembly joining and quality thresholds affect contiguity, error correction, and structurally complex genomic regions. Parameters explored included: Hifiasm default settings, post-joining (joining overlapping contigs to improve contiguity), exclusion of post joining (retaining contigs without merging overlaps), and low quality (lowQ50; removal of contigs with $>50\%$ internal inconsistency as flagged by assembly QC metrics). After initial assembly, all output graph files (.gfa) were parsed with scripts to quantify the distribution of contig lengths, average coverage per contig, and contiguity statistics (total count, total assembly size, average contig length, and N50). Metrics were calculated using QUAST (Gurevich et al. 2013), and single copy ortholog completeness was assessed using the BUSCO squamata_odb12 dataset (Simão et al. 2015). K-mer ($k = 31$) and BUSCO completeness are benchmarks used to examine gene-family gain/loss, with k-mer completeness (although not used in this step) in tandem with assembly contiguity relevant for synteny and structural variation (SV) studies, and BUSCO completeness serving as a criterion using single-copy orthologous genes.

Contig Distribution

Assessment of contig length and coverage profiles in the final assembly revealed a pronounced skew toward smaller, lower-coverage contigs (figure 6). The density distribution of contig lengths peaked approximately at a small 70 kb, likely indicative of either fragmented sequence regions or the presence of unresolved repeats (figure 6). In parallel, the coverage distribution displayed a bimodal structure, with a dominant peak at less than 5× coverage and a secondary, broader peak between 12× and 15× coverage (mean = 7.5×; median = 6.0×) (figure 6). This pattern is indicative of a bulk population of contigs with either suboptimal representation in the sequencing data or regions that are structurally repetitive or difficult to fully resolve (Peona et al. 2019).

Notably, when contig length and coverage were plotted jointly, the analysis revealed a prominent cluster of contigs with coverage between 10× and 15× and lengths below approximately 20 Mb (figure 6). This two-dimensional clustering reflects the genomic architecture wherein high-quality, contiguous regions are more accurately reconstructed, while shorter or less covered contigs may arise from sequence collapse, haplotype divergence, or technical assembly limitations (Kronenberg et al. 2021). The expected distribution profile emphasises the challenge of resolving structurally complex regions, which may be underrepresented or broken across multiple low-quality contigs. Nevertheless, the presence of fewer, longer and more complete contigs in the higher coverage range is a hallmark of successful assembly optimisation. The ‘*no post-join lowQ50*’ de novo assembly was selected to continue with for the assembly as it maintained high BUSCO completeness of 98.97% (with low duplicates at 0.47% and fragments at 0.31%), and an N50 of 28.0 Mb, while its higher contig count (1896) suggests fewer potentially erroneous joins. By avoiding post-joining and incorporating a low-quality threshold (lowq50), it offers a “higher ceiling” (i.e., increasing the quality potential) for improvement in scaffolding, as computational tools will correct or split at potential breakpoints without inherited errors, rather than removing the contigs entirely, ultimately enhancing both accuracy and contiguity in the final assembly (Priyam et al. 2021; Zeng et al. 2023).

Hi-C Guided Pseudo-Chromosome Scaffolding & Manual Curation

The refined de novo assembly was scaffolded to pseudo-chromosome level using HapHiC (Zeng et al., 2024), which iteratively applies Hi-C contact frequencies to order, cluster, and orient contigs using haplotype-aware correction schemes. Eighteen unique parameter combinations were tested, varying Hi-C link weighting, chromosome number specification, mutation probability, population size, number of generations for convergence, cluster separation thresholds, and error correction rounds. Assembly quality from each iteration was assessed using QUAST (Gurevich et al. 2013), Compleasm (Huang & Li 2023), and Merqury v1.4 (Rhie et al., 2020), evaluating N50, contig count, BUSCO completeness, k-mer QV, error rate, and completeness metrics. Pseudo-chromosome assembly ‘*17-hmaj-default-primary.dual-scaff.no-post-join.lowQ50-n14-correct.rounds.3*’ was determined as the optimal combination of contiguity, accuracy and

completeness from the set and hence selected to pass to manual scaffolding. This assembly held the highest N50 (141,678,088) and BUSCO score (99.22%) in addition to the lowest contig count (1229).

Table 1: Comparison of assembly parameters and quality metrics across different Hifiasm configurations.

Assembly	Contig count	Total length	Average coverage	N50	BUSCO %	BUSCO dups %
Primary	1,815	2,166 Mb	8.0	29,195,111	—	—
Default	1,717	2,172 Mb	7.8	37,755,421	98.97	0.47
No Post-join	1,896	2,172.6 Mb	7.5	28,000,000	98.97	0.47
Post-join & LowQ 50	1,717	2,172.5 Mb	7.2	37,800,000	98.97	0.47
Post-join	1,717	2,172.5 Mb	7.2	37,800,000	98.97	0.47
No Post-join & LowQ 50	1,896	2,172.6 Mb	7.5	28,000,000	98.97	0.47

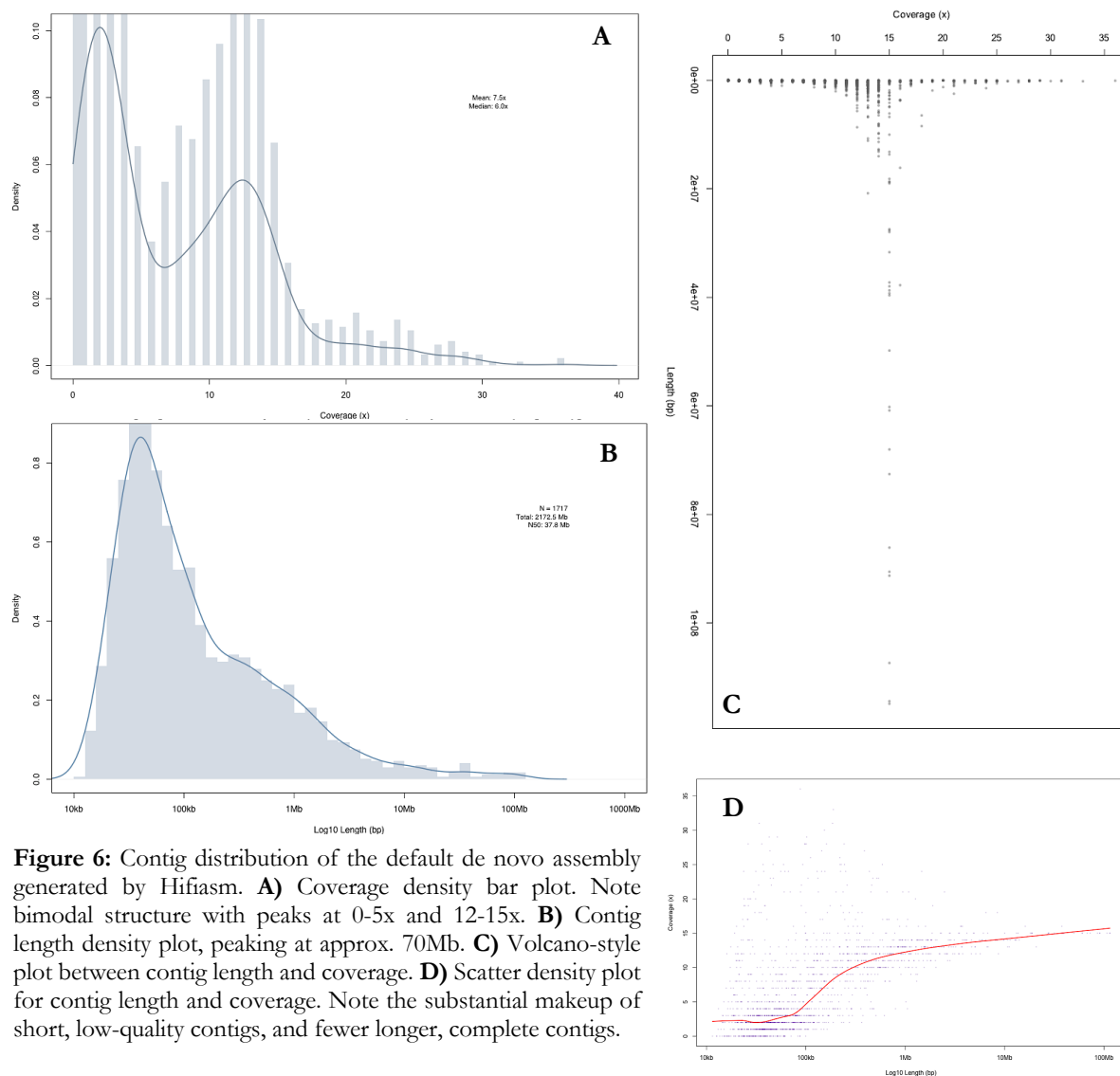


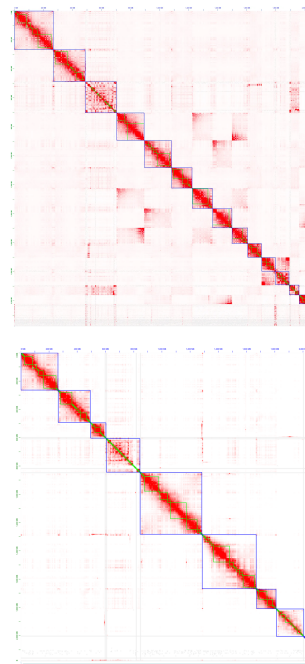
Figure 6: Contig distribution of the default de novo assembly generated by Hifiasm. **A)** Coverage density bar plot. Note bimodal structure with peaks at 0-5x and 12-15x. **B)** Contig length density plot, peaking at approx. 70Mb. **C)** Volcano-style plot between contig length and coverage. **D)** Scatter density plot for contig length and coverage. Note the substantial makeup of short, low-quality contigs, and fewer longer, complete contigs.

Manual scaffolding was then performed using Juicebox v2.25 (Durand et al. 2016) to visually inspect and correct the Hi-C contact matrices. Hi-C heatmaps were loaded to verify structural linearity along each chromosome, with adjustments made to address low-confidence regions or misoriented contigs. Limitations of manual scaffolding are manifested by subjective human judgement, limited resolution of the data and consequent error. This was underscored in the orientation of the Z chromosome, whose lower contact density and lower resolution continued to prove difficult to assemble, illustrated by the manual scaffolding's heatmap (figure 7 (bottom): 5th box from the top left), where there is a relative decrease or

lack of linear red line denoting lower confidence sequence of contigs caused by inherent haploid representation and assembly complexity in the interaction data.

Assembly	Input/Parameter	Contiguity (N50)	BUSCO (%)	BUSCO (D) (%)	K-mer QV	K-mer Error Rate	K-mer Comp (%)	GC (%)	Contigs	Size (bp)
hmap-default	Default parameters (hifiasm de novo assembly)	37,755,421	98.97	0.47	—	—	—	—	1717	2,172Mb
01	Chromosomes = 16	140,834,287	99.21	0.47	62.5484	5.5611e-07	93.6172	40.48	1248	2,172,535,804
02	Chromosomes = 20	131,037,746	99.19	0.47	62.5484	5.5611e-07	93.6172	40.48	1252	2,172,535,804
03	Chromosomes = 16, Mutation Probability = 0.15	140,834,287	99.21	0.47	62.5484	5.5611e-07	93.6172	40.48	1248	2,172,535,804
04	Chromosomes = 20, Mutation Probability = 0.15	131,037,746	99.19	0.47	62.5484	5.5611e-07	93.6172	40.48	1252	2,172,535,804
05	Chromosomes = 16, Max generations for convergence = 10,000	140,834,287	99.21	0.47	62.5484	5.5611e-07	93.6172	40.48	1248	2,172,535,804
06	Chromosomes = 20, Max generations for convergence = 10,000	131,037,746	99.19	0.47	62.5484	5.5611e-07	93.6172	40.48	1252	2,172,535,804
07	Chromosomes = 16, Population size = 200	140,834,287	99.21	0.47	62.5484	5.5611e-07	93.6172	40.48	1248	2,172,535,804
08	Chromosomes = 20, Population size = 200	131,037,746	99.19	0.47	62.5484	5.5611e-07	93.6172	40.48	1252	2,172,535,804
09	Chromosomes = 16, Max rounds for assembly correction = 3	141,678,088	99.21	0.47	62.5484	5.5611e-07	93.6172	40.48	1231	2,172,535,804
10	Chromosomes = 20, Max rounds for assembly correction = 3	107,544,734	99.20	0.47	62.5484	5.5611e-07	93.6172	40.48	1235	2,172,535,804
15	Chromosomes = 14	140,834,287	99.22	0.47	62.5484	5.5611e-07	93.6172	40.48	1246	2,172,535,804
16	Chromosomes = 18	134,635,490	99.21	0.47	62.5484	5.5611e-07	93.6172	40.48	1250	2,172,535,804
17	Chromosomes = 14, Max rounds for assembly correction = 3	141,678,088	99.22	0.47	62.5484	5.5611e-07	93.6172	40.48	1229	2,172,535,804
18	Chromosomes = 18, Max rounds for assembly correction = 3	139,879,401	99.21	0.47	62.5484	5.5611e-07	93.6172	40.48	1233	2,172,535,804

Table 2 (above): Computed quality control metric results from the different combinations of parameters input into HapHiC for scaffolding. Assembly #17 was determined as the superior assembly. **Figure 7 (right):** Heatmaps outputted from JuiceBox Hi-C scaffolding before (top) and after (bottom) manual curation.



The final quality assessment confirmed that deliberate manual scaffolding, informed by Hi-C chromatin interaction data, yielded a further increment in quantitative assembly metrics (BUSCO: 99.22% to 99.24%; total size: 2,172,535,804 bp to 2,172,660,484 bp). This improvement substantiates the value of integrative non-algorithmic curation in the final stabilisation of chromosome-scale assemblies. Importantly, comparison across successive intermediate assemblies clearly demonstrates progressive gains in both completeness and structural accuracy, culminating in a statistically and biologically superior genome assembly.

Evaluation of New Assembly Quality

K-mer analysis

To evaluate the revised assemblies at the sequence level, k-mer spectrum analysis was performed using Merquy, a reference-free evaluation tool that computes quality value (QV), base-level error rate, k-mer completeness, and spectra multiplicity distributions (Rhie et al. 2020). The enhanced assembly showed an increase in base accuracy, with QV improving from 61.51 to 62.55, and a notable reduction in the estimated error rate (from 7.06×10^{-7} to 5.56×10^{-7}). There was a slight decline in overall k-mer completeness (from 95.55% to 93.62%), which most likely reflects the intentional removal of fragmented or error-prone k-mers by computational algorithmic processes such as contig trimming, repeat collapse resolution, assembly error correction, and filtering to reduce redundancy. Biologically, these shifts indicate a more reliable and accurate read set, as cleaner k-mer spectra reflect that a greater proportion of genuine biological variance is now

successfully incorporated into the assembly. This reliability is critical for downstream gene-centric and comparative analyses. Spectra multiplicity plots further support this inference (figure 8): the new assembly's k-mer spectra display a sharper single-copy peak and a narrower distribution. In contrast, the original assembly's spectra are marked by a wider peak and broader tail, indicating higher frequencies of unassigned or erroneous k-mers. Such distributions are consistent with a reduction in artifactual duplications and uncollapsed repeats (Rhie et al. 2020).

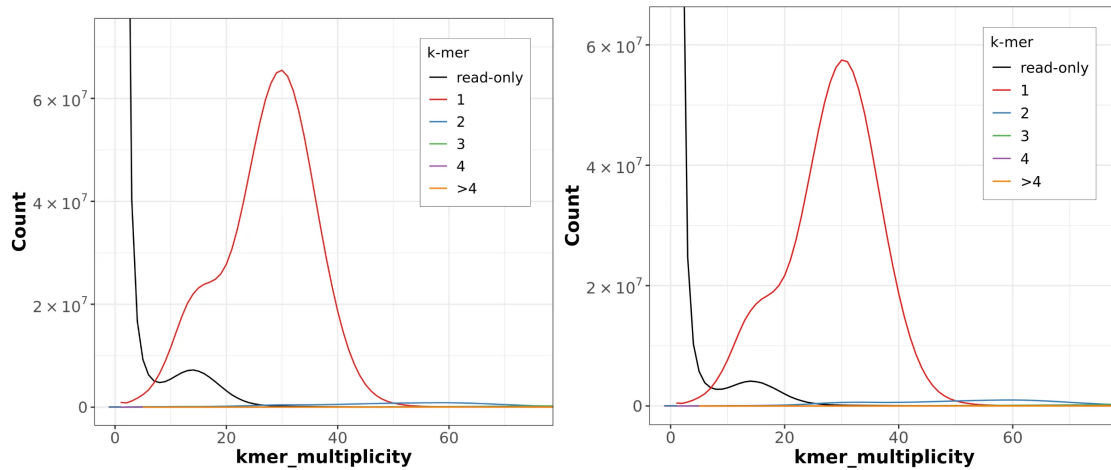


Figure 8 (above): K-mer spectra multiplicity plots outputted from Merquy k-mer analysis. The left plot is from the original assembly, and the right shows results from the revised chromosome-level assembly. Note the new (right) plot has a higher peak, and narrower distribution curve.

Coverage analysis

Genome-wide coverage profiles were recalculated using Mosdepth and graphically compared across assemblies (Pedersen & Quinlan 2018). Focusing on chromosome 2, which previously contained a dense cluster of vomeronasal type-2 receptor (V2R) genes, the new assembly exhibited much greater uniformity and contiguity in coverage, particularly across regions that originally displayed dramatic troughs (figure 9). These changes are consistent with the successful mitigation of reference-biased mapping artifacts and the resolution of technical challenges stemming from high-copy and repetitive content (Wick & Holt 2020).

The iterative refinement of our assembly pipeline yielded a final genome assembly of exceptional quality, demonstrating substantial and critical improvements over the initial reference. While the foundational contiguity, as measured by N50, was already high and remained stable (268.3 Mb to 269.4 Mb, +0.42%), the most significant advancements were observed in metrics of biological completeness and structural accuracy. The implementation of a dedicated Hi-C scaffolding and manual curation pipeline drove the single-copy ortholog completeness (BUSCO) from 96.0% to a near-saturating 99.24%, effectively capturing much of the universal gene space. In parallel, we observed a marked 53% reduction in the BUSCO duplication rate (1.0% to 0.47%), a finding that strongly indicates the successful resolution of previously collapsed heterozygous regions into a haploid representation, thereby reducing artifactual redundancy.

This enhancement in structural fidelity was corroborated by a significant increase in base-level accuracy. The k-mer-based quality value (QV) improved from 61.51 to 62.55, reflecting a 21.25% reduction in the estimated sequencing error rate. The consolidation of the assembly is further evidenced by a 5.91% reduction in total contig count, indicating a more contiguous and less fragmented architecture despite the consistent total genome size.

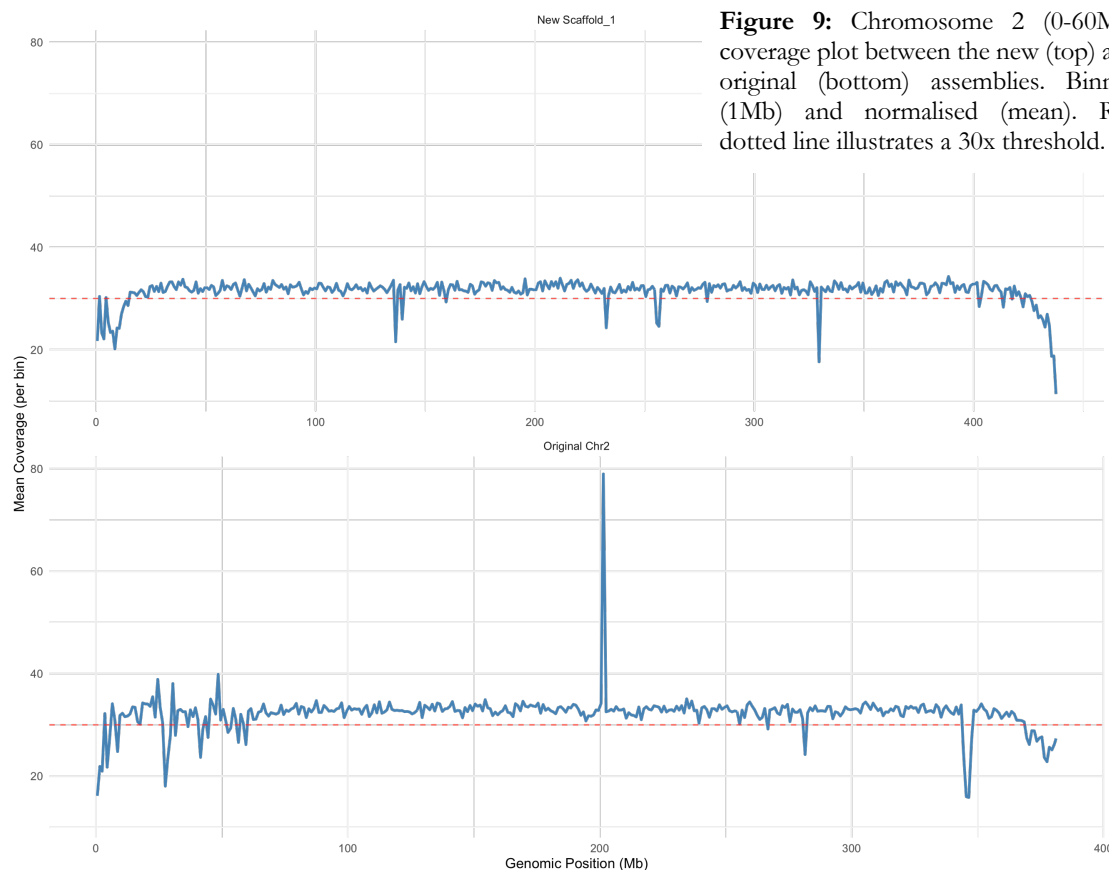


Figure 9: Chromosome 2 (0-60Mb) coverage plot between the new (top) and original (bottom) assemblies. Binned (1Mb) and normalised (mean). Red dotted line illustrates a 30x threshold.

Improved Accuracy of V2R Quantification

In concert, these metrics reduced duplication, elevated completeness, and superior base accuracy, validating improvement in chromosome-level genome assembly of *H. major* (Li et al. 2021). However, although quantification of this enhancement theoretically substantiates for a more robust and reliable resource for subsequent analyses, resolution of the problematic V2R gene annotation was still undetermined.

Chromosome 2 Synteny Analysis

Pairwise synteny analysis between the previous and reconstructed chromosome 2 assemblies was performed using FastGA and visualised using RIdeogram (figure 10) (Hao et al. 2020; Myers, Durbin & Zhou 2025). Overlaying the coordinates of V2R genes revealed that the majority of unsyntenic and inversive regions coincided directly with large V2R clusters (figure 10). This finding strongly supports the hypothesis that the initial assembly errors within these regions, likely due to structural mis joins or collapsed repeat content,

had materially affected annotation accuracy. Such a pattern is highly consistent with phenomena observed in other high-copy, fast-evolving chemoreceptor gene families (Li et al. 2021).

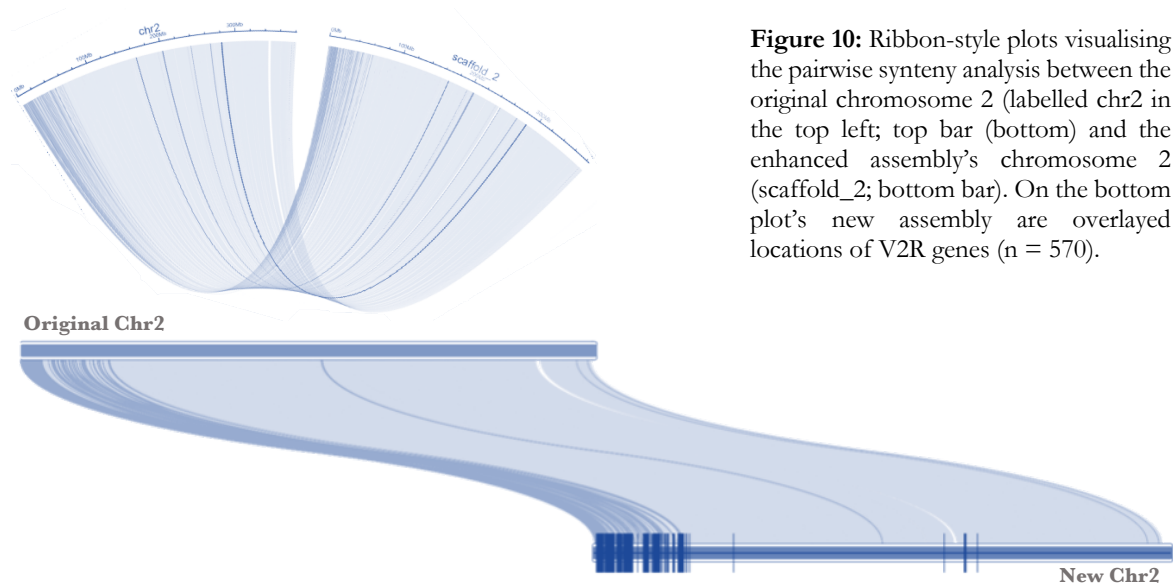


Figure 10: Ribbon-style plots visualising the pairwise synteny analysis between the original chromosome 2 (labelled chr2 in the top left; top bar (bottom) and the enhanced assembly's chromosome 2 (scaffold_2; bottom bar). On the bottom plot's new assembly are overlaid locations of V2R genes ($n = 570$).

Revised V2R annotation & quantification

V2R gene counts were performed across the genome, and for chromosome 2 individually using tBLASTn (BLAST+ v2.13.0) with an E-value threshold of $1e-10$ to identify V2R gene family members using 7-transmembrane domains as a proxy for V2R genes (Camacho et al. 2009; Ludington & Sanders 2025 (unpublished)). Custom databases were created using *makeblastdb*, and results were processed using R/Tidyverse for data analysis (Wickham et al. 2019). Genomic intervals were consolidated with BEDTools *merge* function, and the workflow was managed via SLURM with micromamba for environment control (<https://github.com/mamba-org/mamba>; Quinlan & Hall 2010; Yoo, Jette & Grondona 2003).

Analysis revealed a total of 2,370 putative V2R genes distributed genome-wide in the new assembly, with 570 residing on chromosome 2 and 426 on the Z chromosome. These values are far higher than previously reported for other *Hydrophis* sea snakes. For example, Li et al. (2021) reported 1,411 V2Rs in *H. cyanocinctus* and 1,065 in *H. curtus*. Furthermore, on chromosome 2 alone, Li et al. (2021) recorded 390 V2Rs in *H. curtus*, and the prior *H. major* assembly referenced here contained only 340, both of which are markedly exceeded by the new count of 570 (Ludington & Sanders 2025 (unpublished)). Similarly, the Z chromosome V2R count held 426 in the new assembly, surpassing the 249 reported by Li et al. (2021) for the Z of *H. cyanocinctus*. These results unambiguously demonstrate that systematic improvement of genome sequence and assembly quality translates directly into heightened accuracy and sensitivity of gene family quantification and evolutionary inference.

Discussion

Revised quantification of 2,370 V2R genes across the *H. major* genome represents a substantial upward revision of chemosensory gene repertoire estimates for marine elapids. Li et al.'s (2021) estimate of V2R expansion in *Hydrophis* is significantly underestimated, with the true count in *H. major* being 68% higher than reported for *H. cyanocinctus* and 123% higher than *H. curtus*. This discrepancy reflects both technical limitations of earlier assemblies and genuine biological variation among species. The 570 V2Rs identified on chromosome 2 similarly exceed the 390 V2Rs in *H. curtus* and 340 in the original *H. major* assembly found by Ludington & Sanders (2025 (unpublished)), representing a 42-68% increase attributable to improved resolution of structurally complex regions.

The synteny plot (figure 10) showing resolution of previously problematic regions between chromosome 2 and the Z chromosome suggests that historical assembly artifacts may have conflated or fragmented V2R gene clusters. Therefore, chromosome 2's surge may derive partially from refined demarcation of paralogs formerly entangled with Z chromosome sequences, as substantiated by the synteny analysis resolving superimposed inversions (figure 10). Li et al. (2021) and Ludington et al. (2023) had similar findings following chromosome 2 synteny analysis (figure 1 & 2), characterised by poor collinearity across the V2R-rich region of interest, yet broad synteny across the remaining chromosome length. This pattern indicates that assembly artifacts in repeat-dense, structurally variable regions have systematically biased previous estimates of chemosensory gene repertoire size.

Several technical and biological factors contribute to these numerical discrepancies. The technological advancement from PacBio continuous long reads (CLR) to PacBio high-fidelity (HiFi) reads likely accounts for a proportion of the increased gene counts. Li et al. (2021) used PacBio Sequel CLR sequencing with approximately 85-90% raw read accuracy, while the *H. major* assembly employed PacBio Sequel II HiFi reads with >99% base accuracy (Q20+) (Ludington et al. 2023). The accuracy of HiFi reads is critical for properly resolving tandemly duplicated V2R arrays that may collapse in lower-accuracy long-read assemblies, as similar paralogous copies can be distinguished with greater confidence. New V2R repertoire counts, and composition warrant consideration of the extant biological and evolutionary inferences drawn from the underestimated data.

Repeat Elements & Genome Structural Variation

The new chromosome-scale assemblies for *Hydrophis* species have altered our understanding of their evolutionary implications, including speciation mechanics and radiative adaptation, by providing resolution of structural variants (SVs) and gene clusters previously obscured in fragmented genomes (Ludington et al., 2023; Li et al., 2021). The lack of positively selected genes in SV-dense zones, now clearly delineated, bolsters models where architectural barriers, rather than directional selection, perpetuate diversity in early divergence (Cruickshank & Hahn 2014; Ludington et al. 2023). SV predominance at chromosomal

extremities, with copious inversions, suppresses recombination and erects genetic impediments, as revealed by enhanced synteny analyses (Ludington et al., 2023). These high-quality assemblies also highlight how SVs in clusters sustain functional multiplicity under balancing selection, fuelling adaptive radiation by preserving chemosensory diversity (Kishida et al., 2019).

The temporal overlap between long terminal repeat (LTR) retrotransposon expansion and *Hydrophis* radiation strongly suggests that repeat element activity catalysed major genomic reorganisation during the critical marine adaptation and ecological diversification phase. Ludington et al. (2023) report LTR insertions began ~12.5 million years ago, peaking between 2.5 and 5 million years ago, concurrent with *Hydrophis*'s rapid radiation period (3–8 mya; Lee et al. 2016). Shared insertion timing and distribution across four newly assembled sea snake genomes indicate these expansions originated in a common ancestral *Hydrophis* lineage. Critically, copy number variants (CNVs) concentrate within LTR sequences, providing a mechanism for TE expansion to generate SV in V2R-rich regions (Li et al. 2021). Li et al. (2021) noted that CNVs "mostly falling within or across LTRs" highlight 'TEs' roles in facilitating SVs and regulating gene expression by altering cis-regulatory elements. This aligns with broader models of TE-mediated regulatory evolution, where repeats contribute sequences co-opted for novel expression patterns (Bourque et al. 2008; Chuong et al. 2017). Repeat-driven SVs may reposition cis-regulatory elements, altering V2R expression in a lineage-specific manner, a hypothesis supported by observed genomic architecture, though empirical validation is needed. Li et al. (2021) identified 52 balanced SVs disrupting V2R/V1R genic regions and 39 CNVs affecting exons, showing SVs impact gene structure alongside regulatory landscapes.

Terrestrial elapids contrast marine V2R trajectories, with fewer V2Rs in total (generally hundreds) and on chromosome 2 (279 in *Crotalus adamanteus*, 28 in *Thamnophis elegans*), now overshadowed by *Hydrophiinae* totals (Ludington et al. 2023). This gap reflects the intensified selective pressures to adapt chemosensory systems for the marine environment, where chemical cue discrimination is vital for prey detection and social interaction (Crowe-Riddell et al. 2016). Conserved chromosomal synteny alongside lineage-specific SV and TE activity illustrate how repetitive elements shape structural and regulatory genome architecture, facilitating the explosive diversification of V2Rs and underpinning rapid speciation. Relatively conserved synteny across sea snakes, rattlesnakes, and garter snakes (chromosomes 1, 2, Z) indicates ancestral frameworks, but variable copy numbers reflect lineage-specific duplications propelled by TEs like LTRs, fostering SVs and expression modulation (Bohne et al. 2008; Kazazian 2004; Ludington et al. 2023). In *Hydrophiinae* TE-enriched chromosome termini (e.g., chromosome 2) exacerbated this, yielding SV "tangles" lacking positively selected genes but rich in adaptive potential (Ludington et al. 2023).

Evolutionary and Functional Implications of V2R Expansion

This enhanced contiguity refining V2R counts, suggests extraordinary selective pressure to preserve chemosensory diversity during the elapid land-to-sea transition. Each sensory neuron expresses one or few

V2Rs, with repertoires proportional to odorant variety via a combinatorial receptor code (Malnic et al. 1999; Saito et al. 2009). Thus, the amplified repertoire likely enables broader detection of waterborne cues than previously presumed, facilitating radiation into diverse trophic niches in marine environments.

Evolutionary ramifications recast the *Hydrophiinae* diversification paradigm, indicating a more accentuated expansion than previously surmised. V2Rs mediate prey localisation and conspecific recognition via tongue-flicking, crucial in longer-wavelength-dominated aquatic environments (Daghfous et al. 2012; Halpern & Kubie 1983; Li et al. 2021; Ludington et al. 2023). Aggregation in inversion-susceptible loci (~0-60 Mb on chromosome 2) aligns with balancing selection signatures across *Hydrophis* taxa (Ludington & Sanders 2025 (unpublished)), sustaining ancestral polymorphisms to expedite responses to emergent niches or fortify isolation amid rapid radiation (~3–8 Ma). Li et al. (2021) linked expansive inversions (up to 90 V2Rs) to dietary schisms: *H. cyanocinctus* favouring anguilliform prey versus *H. curtus*' generalist across benthic, demersal, and pelagic fish. The assembly's precision intensifies this and similar narratives, postulating SV-induced recombination suppression propelled ecological segregation and speciation. SV mapping fracturing genic domains or pathways (e.g., CNVs in *ITPR3*, *TRPC2*), suggesting they underpin prey selectivity and mitigate hybridisation in near-simultaneous divergence of sympatric species (Li et al. 2021; Ludington & Sanders, 2025; Welleneruther et al. 2019). This underscores the assembly's relevance for functional studies linking architecture to ecological success.

Importance of Parametrisation & Strategic Fragmentation

The results propose that applying initially liberal and fragmentation-permissive parameters early (i.e., during *de novo* assembly) elevates the maximum attainable structural accuracy by preventing premature contig joins, thus producing smaller but more reliable contigs. These contigs serve as a high-fidelity substrate that can be accurately scaffolded in subsequent steps with complementary evidence and long-range data such as Hi-C interaction maps. Completion of additional assemblies applying this workflow could provide further corroboration of this principle, supporting the paradigm that early deliberate parameter tuning unlocks assembly potential and improves biological relevance. Priyam et al. (2021) demonstrated that systematic fine-tuning of assembly parameters considerably improves assembly quality, stating that varying thresholds for locating read overlaps improves contiguity, completeness, and accuracy of assemblies, with 64% of explored parameters outperforming default settings. Comparably, recent developments by Zeng et al. (2023) in haplotype-aware Hi-C scaffolding uncover that initial fragmentation under relaxed parameters yield statistically more contiguous final assemblies. Together, these findings align with this study's results, punctuating controlled fragmentation paired with orthogonal scaffolding evidence as a tool to optimise genome assembly continuity and biological fidelity.

Conclusion

Here, it is demonstrated that systematic parameter optimisation in genome assembly pipelines can profoundly augment structural accuracy and gene quantification in non-model organisms, as evidenced by the refined *Hydrophis major* chromosome-scale assembly. By prioritising iterative quality assessment and liberal parameterisation, rather than default settings, this research delivered a chromosome-level assembly with substantive completeness, accuracy, and hence reliability, resulting in resolved artifacts in target regions, and yielding an enhanced V2R gene catalogue. These improvements refine our understanding of chemosensory adaptation in marine elapids, revealing how transposable element expansions and structural variants underpin rapid diversification and ecological specialisation.

Integrating Hi-C for 3D mapping could test how inversions and LTR-driven SVs rewire enhancer-promoter interactions, creating regulatory modules akin to mammalian olfactory clusters (Monahan et al., 2019). Tissue-specific RNA-seq, ATAC-seq, and expression profiling across ontogeny, sexes, and niches would correlate structural variation with prey preferences and behavioural contexts, validating modularity hypotheses (Shih et al., 2016). The premise of embracing deliberate, data-informed parameter exploration over generic defaults or reliance on commonplace inputs, is pivotal for unlocking assembly potential and advancing genomic resources – future genomics must welcome such rigor to unravel the architecture of evolution.

Reference List

- Böhne, A., Brunet, F., Galiana-Arnoux, D., Schultheis, C. & Volff, J.-N. 2008, 'Transposable elements as drivers of genomic and biological diversity in vertebrates', *Chromosome Research*, vol. 16, no. 1, pp. 203–215.
- Bourque, G., Burns, K.H., Gehring, M., Gorbunova, V., Seluanov, A., Hammell, M., Imbeault, M., Izsvák, Z., Levin, H.L., Macfarlan, T.S., Mager, D.L. & Feschotte, C. 2018, 'Ten things you should know about transposable elements', *Genome Biology*, vol. 19, no. 199. Available at: <https://genomebiology.biomedcentral.com/article/s/10.1186/s13059-018-1577-z> (Accessed: 7 November 2025).
- Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K. & Madden, T.L. 2009, 'BLAST+: architecture and applications', *BMC Bioinformatics*, vol. 10, no. 1, p. 421.
- Cheng, H., Concepcion, G.T., Feng, X., Zhang, H. & Li, H. 2021, 'Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm', *Nature Methods*, vol. 18, no. 2, pp. 170–175.
- Chuong, E.B., Elde, N.C. & Feschotte, C. 2017, 'Regulatory activities of transposable elements: from conflicts to benefits', *Nature Reviews Genetics*, vol. 18, no. 2, pp. 71–86.
- Crowe-Riddell, J.M., Snelling, E.P., Watson, A.P., Suh, A.K., Partridge, J.C. & Sanders, K.L. 2016, 'The evolution of scale sensilla in the transition from land to sea in elapid snakes', *Open Biology*, vol. 6, no. 6, p. 160054.
- Daghfous, G., Smargiassi, M., Libourel, P.-A., Wattiez, R. & Bels, V. 2012, 'The function of oscillatory tongue-flicks in snakes: insights from kinematics of tongue-flicking in the banded water snake (*Nerodia fasciata*)', *Chemical Senses*, vol. 37, no. 9, pp. 883–896.
- Denton, J.F., Lugo-Martinez, J., Tucker, A.E., Schrider, D.R., Warren, W.C. & Hahn, M.W. 2014, 'Extensive error in the number of genes inferred from draft genome assemblies', *PLoS Computational Biology*, vol. 10, no. 12, p. e1003998.
- Durand, N.C., Shamim, M.S., Machol, I., Rao, S.S.P., Huntley, M.H., Lander, E.S. & Aiden, E.L. 2016, 'Juicer provides a one-click system for analyzing loop-resolution Hi-C experiments', *Cell Systems*, vol. 3, no. 1, pp. 95–98. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5846465/> (Accessed: 26 February 2021).
- Gurevich, A., Saveliev, V., Vyahhi, N. & Tesler, G. 2013, 'QUAST: quality assessment tool for genome assemblies', *Bioinformatics*, vol. 29, no. 8, pp. 1072–1075.
- Halpern, M. & Kubie, J.L. 1983, 'Snake tongue flicking behavior: clues to vomeronasal system functions', in D. Müller-Schwarze & R.M. Silverstein (eds), *Chemical Signals in Vertebrates 3*, Springer, Boston, MA, pp. 45–72.
- Hao, Z., Lv, D., Ge, Y., Shi, J., Weijers, D., Yu, G. & Chen, J. 2020, 'RIdiogram: drawing SVG graphics to visualize and map genome-wide data on the idiograms', *PeerJ Computer Science*, vol. 6, p. e251.
- Huang, N. & Li, H. 2023, 'compleasm: a faster and more accurate reimplement of BUSCO', *Bioinformatics*, vol. 39, no. 10. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10558035/> (Accessed: 22 December 2023).
- *Hydrophis major* genome assembly HMAJ_1.0 2025, National Center for Biotechnology Information. Available at: https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_033807585.1/ (Accessed: 7 November 2025).
- Kazazian, H.H. 2004, 'Mobile elements: drivers of genome evolution', *Science*, vol. 303, no. 5664, pp. 1626–1632.
- Kronenberg, Z.N., Rhie, A., Koren, S., Concepcion, G.T., Peluso, P., Munson, K.M., Porubsky, D., Kuhn, K., Mueller, K.A., Low, W.Y., Hiendleder, S., Fedrigo, O., Liachko, I., Hall, R.J., Phillippy, A.M., Eichler, E.E., Williams, J.L., Smith, T.P.L., Jarvis, E.D. & Sullivan, S.T. 2021, 'Extended haplotype-phasing of long-read de novo genome assemblies using Hi-C', *Nature Communications*, vol. 12, no. 1, p. 1935.
- Li, A., Wang, J., Sun, K., Wang, S., Zhao, X., Wang, T., Xiong, L., Xu, W., Qiu, L., Shang, Y., Liu, R., Wang, S. & Lu, Y. 2021, 'Two reference-quality sea snake genomes reveal their divergent evolution of adaptive traits and venom systems', *Molecular Biology and Evolution*, vol. 38, no. 11, pp. 4867–4881.
- Li, H. 2018, 'Minimap2: pairwise alignment for nucleotide sequences', *Bioinformatics*, vol. 34, no. 18, pp. 3094–3100.
- Ludington, A.J., Hammond, J.M., Breen, J., Deveson, I.W. & Sanders, K.L. 2023, 'New chromosome-scale genomes provide insights into marine adaptations of sea snakes (Hydrophis: Elapidae)', *BMC Biology*, vol. 21, no. 1, p. 39.
- Malnic, B., Hirono, J., Sato, T. & Buck, L.B. 1999, 'Combinatorial receptor codes for odors', *Cell*, vol. 96, no. 5, pp. 713–723.
- Myers, G., Durbin, R. & Zhou, C. 2025, 'FastGA: Fast Genome Alignment', *Bioinformatics Advances*, vol. 5, no. 1, p. vbae100.
- Pedersen, B.S. & Quinlan, A.R. 2018, 'Mosdepth: quick coverage calculation for genomes and exomes', *Bioinformatics*, vol. 34, no. 5, pp. 867–868.
- Peona, V., Blom, M.P.K., Xu, L., Burri, R., Sullivan, S., Bunikis, I., Liachko, I., Jönsson, K.A., Zhou, Q., Irestedt, M. & Suh, A. 2019, 'Identifying the causes and consequences of assembly gaps using a multiplatform genome assembly of a bird-of-paradise', *Molecular Ecology Resources*, vol. 21, no. 1, pp. 263–286.
- Priyam, A., Witwicka, A., Brahma, A., Stolle, E. & Wurm, Y. 2021, *Parameter exploration improves the accuracy of long-read genome assembly*, Queen Mary University of London. Available at: <https://wurmlab.com/publications/parameter-exploration-improves-genome-assembly-accuracy/> (Accessed: 7 November 2025).
- Quinlan, A.R. & Hall, I.M. 2010, 'BEDTools: a flexible suite of utilities for comparing genomic features', *Bioinformatics*, vol. 26, no. 6, pp. 841–842.
- Rhie, A., Walenz, B.P., Koren, S. & Phillippy, A.M. 2020, 'Mercury: reference-free quality, completeness, and phasing assessment for genome assemblies', *Genome Biology*, vol. 21, no. 1, p. 245.

- Robinson, J.T., Thorvaldsdóttir, H., Winckler, W., Guttman, M., Lander, E.S., Getz, G. & Mesirov, J.P. 2011, 'Integrative genomics viewer', *Nature Biotechnology*, vol. 29, no. 1, pp. 24–26.
- Saito, H., Chi, Q., Zhuang, H., Matsunami, H. & Mainland, J.D. 2009, 'Odor coding by a mammalian receptor repertoire', *Science Signaling*, vol. 2, no. 60, p. ra9.
- Simão, F.A., Waterhouse, R.M., Ioannidis, P., Kriventseva, E.V. & Zdobnov, E.M. 2015, 'BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs', *Bioinformatics*, vol. 31, no. 19, pp. 3210–3212.
- Sutton, J.M., Millwood, J.D., Case McCormack, A. & Fierst, J.L. 2021, 'Optimizing experimental design for genome sequencing and assembly with Oxford Nanopore Technologies', *GigaByte*, vol. 2021. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9650304/> (Accessed: 7 November 2025).
- Wellenreuther, M., Mérot, C., Berdan, E. & Bernatchez, L. 2019, 'Going beyond SNPs: the role of structural genomic variants in adaptive evolution and species diversification', *Molecular Ecology*, vol. 28, no. 6, pp. 1203–1209.
- Wick, R.R. & Holt, K.E. 2020, 'Benchmarking of long-read assemblers for prokaryote whole genome sequencing', *F1000Research*, vol. 8, p. 2138.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V. & Takahashi, K. 2019, 'Welcome to the Tidyverse', *Journal of Open Source Software*, vol. 4, no. 43, p. 1686.
- Yoo, A.B., Jette, M.A. & Grondona, M. 2003, 'SLURM: Simple Linux Utility for Resource Management', in D. Feitelson, L. Rudolph & U. Schwiegelshohn (eds), *Job Scheduling Strategies for Parallel Processing*, Springer, Berlin, Heidelberg, pp. 44–60.
- Zeng, X., Yi, Z., Zhang, X., Du, Y., Li, Y., Zhou, Z., Chen, S., Zhao, H., Yang, S., Wang, Y. & Chen, G. 2024, 'Chromosome-level scaffolding of haplotype-resolved assemblies using Hi-C data without reference genomes', *Nature Plants*, vol. 10, no. 8, pp. 1184–1200.